

Practical Demo: Duration, Convexity, and Portfolio Repricing

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

A manager holds three cash bonds and asks: **if yields rise by 50 basis points, how much will we lose, and can a duration estimate be trusted?** Start with the payments you

own, then measure risk.

Purpose and Assumptions

This demo prices a small bond portfolio from contractual cash flows, measures Macaulay duration, modified duration, DV01/PVBP, and convexity, and compares approximations with exact repricing. All bonds and yields are **illustrative, market-like inputs**, not live quotes. Bonds pay semiannual coupons; yields are nominal annual yields compounded semiannually; settlement is on a coupon date; default, taxes, accrued interest, and embedded options are ignored.

Workflow stage	In this case
Observed on a position report	Face value, coupon, maturity, and yield
Calculated	Full price, duration, DV01, and convexity
Checked	Approximation error against exact cash-flow repricing
Used for	Portfolio shock P&L and identification of curve risk

1 Full Cash-Flow Pricing and Risk Measures

F = face value in dollars, c = annual coupon rate,
 T = maturity in years, y = nominal annual yield,
 $m = 2$ payments per year.

$$n = mT, \quad t_k = \frac{k}{m}, \quad CF_k = \frac{Fc}{m} + F\mathbf{1}_{\{k=n\}}, \quad k = 1, \dots, n.$$

$$PV_k = \frac{CF_k}{(1 + y/m)^k}, \quad P = \sum_{k=1}^n PV_k.$$

$$D_{\text{Mac}} = \frac{\sum_{k=1}^n t_k PV_k}{P}, \quad D_{\text{mod}} = \frac{D_{\text{Mac}}}{1 + y/m} = -\frac{1}{P} \frac{\partial P}{\partial y}.$$

$$DV01 = PVBP \approx PD_{\text{mod}} \times 10^{-4} \quad (\text{dollars per basis point}).$$

$$C = \frac{1}{P} \frac{\partial^2 P}{\partial y^2} = \frac{1}{P} \sum_{k=1}^n \frac{k(k+1)CF_k}{m^2(1+y/m)^{k+2}}.$$

Bond	2-year Treasury	5-year agency	10-year corporate
Face value (\$mm)	12.0000	10.0000	8.0000
Coupon rate	0.0400	0.0475	0.0575
Maturity (years)	2.0000	5.0000	10.0000
Yield	0.0410	0.0485	0.0610
Price (\$)	11,977,181.3216	9,956,069.5998	7,792,674.9472
Macaulay duration	1.9419	4.5077	7.7068
Modified duration	1.9029	4.4010	7.4787
DV01/PVBP (\$/bp)	2,279.1012	4,381.6687	5,827.9378
Convexity	4.6154	22.7992	69.2981

2 Exact Repricing Versus Approximations

i Follow one payment

The corporate bond pays \$230,000 every six months: $\$8,000,000 \times 5.75\% / 2$. Its first payment is worth $\$230,000 / 1.0305$ today, while its final payment is discounted twenty times. Later payments are more sensitive because a yield change affects more discounting periods. This is why maturity alone does not describe the portfolio's interest-rate risk.

For a yield shock Δy ,

$$\frac{\Delta P}{P} \approx -D_{mod} \Delta y$$

and the duration-plus-convexity approximation is

$$\frac{\Delta P}{P} \approx -D_{mod} \Delta y + \frac{1}{2} C (\Delta y)^2.$$



Figure 1: Present value by payment date for the corporate position; the final bar includes principal.

2.1 Explore the shock size (HTML)

Move the slider to shock the **10-year corporate position**: \$8 million face, 5.75% coupon, semiannual payments, and 6.10% initial yield. All three estimates use the same starting price and cash flows. Section 2.2 provides the PDF fallback.

Results retain six decimals in percentage units. Error is **estimated minus exact**, reported in basis points of the initial price, not basis points of yield.

2.2 Approximation table: precision and error units

The corporate position is valued at its original 6.10% yield before every shock. A **+25 bp yield shock** means $6.10\% + 0.25\% = 6.35\%$.

$$r_{\text{exact}} = \frac{P(y + \Delta y) - P(y)}{P(y)}.$$

$$e_{\text{price bp}} = 10,000(r_{\text{estimate}} - r_{\text{exact}}), \quad e_{\$} = P(y)(r_{\text{estimate}} - r_{\text{exact}}).$$

One basis point **of price** equals 0.01 percentage points of price change. Equal values rounded to three decimals need not have zero error. Positive error means the estimated ending value

is too high; negative error means it is too low. All errors below use unrounded calculation results.

Yield shock (bp)	Duration-only change (%)	Duration + convexity change (%)	Exact price change (%)
-150	11.218108	11.997711	12.039028
-75	5.609054	5.803955	5.809013
-25	1.869685	1.891340	1.891525
25	-1.869685	-1.848029	-1.848211
75	-5.609054	-5.414153	-5.419010
150	-11.218108	-10.438504	-10.476594

Yield shock (bp)	Duration-only error (bp of price)	Duration + convexity error (bp of price)
-150	-82.091998	-4.131676
-75	-19.995930	-0.505849
-25	-2.184045	-0.018481
25	-2.147333	0.018232
75	-19.004384	0.485697
150	-74.151333	3.808989

i Why the +25 bp row has a nonzero error

The duration-plus-convexity change is **−1.848029%** and the exact change is **−1.848211%**. Both round to **−1.848%** at three decimals, but

$$10,000[-0.018480289632 - (-0.018482112824)] \approx 0.018232 \text{ bp of price.}$$

Multiplying the unrounded return difference by the initial position value, \$7,792,674.95, gives a **+\$14.21 error**. The approximation understates the loss by \$14.21. Display rounding explains the apparent inconsistency.

3 Portfolio Aggregation

Each column below is a position; the last column reconciles the whole portfolio. Market values and DV01s **sum**. Durations and convexities are **market-value-weighted averages**. Contribution rows show the pieces of those averages.

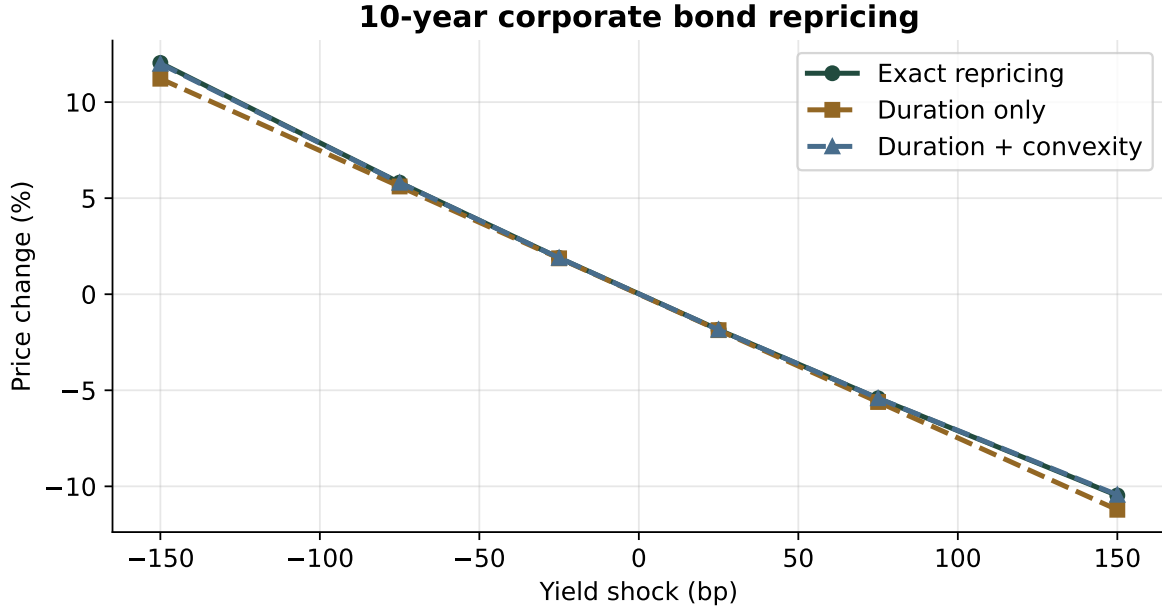


Figure 2: Convexity materially improves the approximation for larger yield moves

$$V_0 = \sum_i P_i, \quad w_i = \frac{P_i}{V_0}, \quad DV01_{\text{port}} = \sum_i DV01_i.$$

$$D_{\text{port}} = \sum_i w_i D_i = \frac{DV01_{\text{port}}}{V_0 10^{-4}}, \quad C_{\text{port}} = \sum_i w_i C_i.$$

	2Y Treasury	5Y agency	10Y corporate	Portfolio
Face value (\$mm)	12.0000	10.0000	8.0000	30.0000
Market value (\$mm)	11.9772	9.9561	7.7927	29.7259
Market-value weight (%)	40.2920	33.4929	26.2151	100.0000
Macaulay duration (years)	1.9419	4.5077	7.7068	4.3125
Modified duration	1.9029	4.4010	7.4787	4.2013
Contribution to duration	0.7667	1.4740	1.9606	4.2013
DV01 (\$/bp)	2,279.1012	4,381.6687	5,827.9378	12,488.7076
Share of DV01 (%)	18.2493	35.0850	46.6657	100.0000
Convexity	4.6154	22.7992	69.2981	27.6623
Contribution to convexity	1.8596	7.6361	18.1665	27.6623

The portfolio's Macaulay duration is the weighted payment time using each position's own discounted cash flows. With different bond yields, do not convert it to portfolio modified duration using a simple average yield. Aggregate each bond's modified duration as shown above.

4 Parallel and Nonparallel Scenarios

A single duration number is designed for a common yield shift. Here we approximate a curve twist with different yield-to-maturity shocks for the three bonds. Repricing is exact for those assigned bond yields, not for a fully specified spot-curve twist. A production key-rate analysis would shock the underlying discount curve and revalue every dated cash flow.

For bond i , let b_i be its assigned shock in basis points, so $\Delta y_i = b_i/10,000$. Each method starts from the **same dollar market value** P_i , not face value. Here D_i is modified duration and C_i is annualized convexity, both evaluated at the bond's initial yield.

$$L_i^D = -P_i D_i \Delta y_i = -DV01_i b_i, \quad A_i^C = \frac{1}{2} P_i C_i (\Delta y_i)^2.$$

$$L_i^{D+C} = L_i^D + A_i^C, \quad L_i^{\text{exact}} = \sum_{k=1}^{n_i} \frac{CF_{i,k}}{[1 + (y_i + \Delta y_i)/m]^k} - P_i.$$

$$L_{\text{port}}^D = \sum_i L_i^D, \quad L_{\text{port}}^{D+C} = \sum_i L_i^{D+C}, \quad L_{\text{port}}^{\text{exact}} = \sum_i L_i^{\text{exact}}.$$

Under the parallel +50 bp shock, every $\Delta y_i = 0.005$, so the same results can also be obtained directly from the portfolio measures:

$$L_{\text{port}}^D = -50 DV01_{\text{port}}, \quad L_{\text{port}}^{D+C} = -V_0 D_{\text{port}} (0.005) + \frac{1}{2} V_0 C_{\text{port}} (0.005)^2.$$

For the twist scenarios, use each bond's own shock **before summing**. There is no common Δy to insert into the portfolio-duration formula.

Bond	Parallel +50 bp	Bear steepener	Bull flattener
2-year Treasury	50	10	-15
5-year agency	50	35	-45
10-year corporate	50	75	-90

Scenario	Duration P&L (\$)	Duration + convexity P&L (\$)	Exact P&L (\$)
Parallel +50 bp	-624,435.38	-614,156.79	-614,300.03
Bear steepener	-613,244.75	-596,638.80	-597,026.60
Bull flattener	755,876.01	780,107.18	780,811.22

5 Validation Checks

Validation asks two different questions: **do the calculations reconcile?** and **does the risk allocation make economic sense?**

The chart compares the share of money invested with the share of sensitivity to a common 1 bp yield increase:

$$\text{Capital share}_i = 100 \frac{P_i}{V_0}, \quad \text{DV01 share}_i = 100 \frac{DV01_i}{DV01_{\text{port}}}.$$

Each set of shares must sum to 100%. If a bond's DV01 bar is longer than its market-value bar, it contributes more risk per dollar invested than the portfolio average. The long corporate bond can therefore dominate rate risk without being the largest investment. This chart diagnoses concentration; it does not by itself validate the pricing formula or measure credit risk.

Predict, change, explain

1. **Predict:** before moving the Section 2.1 slider, write down whether a +150 bp move should produce six times the +25 bp percentage loss. Duration predicts this scaling; curvature should make the actual loss smaller.
2. **Change:** compare +25, +150, and −150 bp. Record all three percentage estimates and the error in bp of price. Keep the bond, face value, and starting yield unchanged.
3. **Explain:** compare the absolute gain at −150 bp with the loss at +150 bp. Positive convexity makes the gain larger. Explain why the combined estimate is closer, but still not exact for a finite shock.
4. **Apply:** use the DV01-share chart to choose a position to reduce risk. Compare DV01 removed per \$1 million face sold, rather than selecting the largest face value automatically. Also consider the desired maturity and credit mix.

The checks below compare independently constructed quantities where possible: a centered finite-difference derivative against analytical DV01, position totals against weighted portfolio

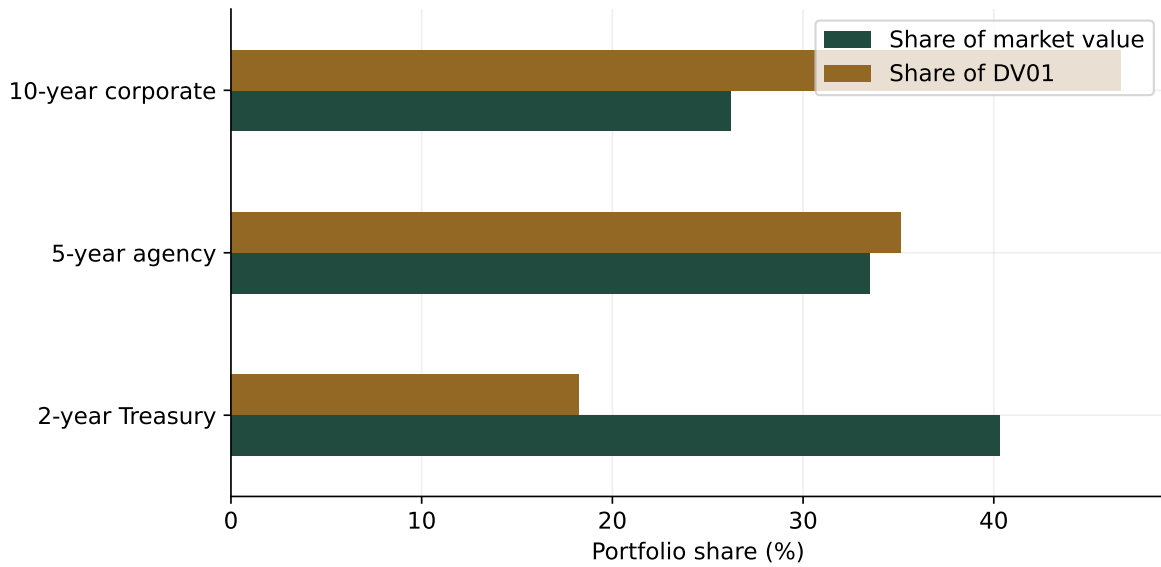


Figure 3: Capital allocation and risk allocation answer different questions.

measures, and exact repricing against the approximations. The duration ordering assumes the positive yields in this case. Passing these tests checks this teaching model, not real-world settlement, option exercise, or credit-spread dynamics.

Passed	
Check	
Positive yields: Macaulay > modified	True
DV01 agrees with centered repricing	True
Value and risk shares each sum to 100%	True
Weighted duration reconciles to total DV01	True
Parallel duration P&L equals -50 x DV01	True
Convexity improves every tabulated shock	True
Exact price falls as yield rises	True

6 Financial Interpretation

Duration is a local slope, convexity is a curvature correction, and exact cash-flow repricing is the benchmark. DV01 translates percentage sensitivity into dollars. Portfolio aggregation is reliable for a stated parallel shock, but nonparallel scenarios require maturity-specific or key-rate exposures.

7 Interactive explorer

Start at a zero shock, then compare +150 and -150 bp. The straight-line estimate has no curvature. Explain which approximation is closest and why equal yield moves do not produce equal percentage gains and losses.

8 Mini Project

Project brief. You are the junior fixed-income analyst supporting a manager of the three-position portfolio in Section 1. Before a rate-risk meeting, the manager wants a reproducible **risk memo and supporting tables**, rather than a list of model outputs. Use the embedded, illustrative positions as the required dataset. Treat the valuation instant as a coupon date for each bond.

Answer: **What is the portfolio worth, how much does it lose under a parallel +50 bp move, and what does a bear steepener reveal that aggregate duration misses?** Recommend one explicitly sized risk-reduction action and discuss its tradeoffs. This is an instantaneous rate-risk exercise, not a forecast of total return.

Required deliverable:

1. A data-and-conventions statement and position table with face value, coupon, maturity, and initial yield. Distinguish supplied inputs from calculated prices.
2. A position-and-portfolio risk exhibit showing market value, weights, duration, DV01, convexity, and each position's contributions.
3. A scenario exhibit with duration, duration-plus-convexity, and exact dollar P&L for +50 bp and the bear steepener, including both signed dollar errors. Include the bull flattener as an opposite-direction check.
4. One worked bond-level calculation, a concentration diagnosis, and a quantified action. State what the action gives up and what risks the model leaves out.
5. A short control statement confirming reconciliation and repricing checks.

Worked deliverable: portfolio rate-risk memo and supporting tables

Decision summary. The portfolio has \$29,725,925.87 of market value against \$30 million face value. DV01 is \$12,488.71, modified duration is 4.201285, and convexity is 27.662301. A parallel +50 bp move produces exact P&L of \$-614,300.03 (-2.066546% of initial value). The bear steepener produces \$-597,026.60. The nonparallel result depends on where the exposure lies, not just total DV01.

Data and conventions. Coupons are semiannual, yields are annual nominal rates compounded twice yearly, and settlement is on coupon dates. Full price equals clean price here because accrued interest is zero. Face value is contractual principal; market value is the discounted value calculated from all remaining payments. No calendar valuation date is assigned.

Exhibit 1. Supplied position inputs

Bond	2-year Treasury	5-year agency	10-year corporate
Face value (\$mm)	12.0000	10.0000	8.0000
Coupon rate	4.00%	4.75%	5.75%
Maturity (years)	2	5	10
Yield	4.10%	4.85%	6.10%

Exhibit 2. Calculated position risks and portfolio reconciliation

	2Y Treasury	5Y agency	10Y corporate	Portfolio
Face value (\$mm)	12.0000	10.0000	8.0000	30.0000
Market value (\$mm)	11.9772	9.9561	7.7927	29.7259
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Exhibit 3. Scenario P&L and signed errors. Errors equal estimated P&L minus exact P&L. Negative error overstates the loss (or understates the gain); positive error overstates the ending value. Bond shocks, ordered Treasury / agency / corporate, are +50 / +50 / +50 bp, +10 / +35 / +75 bp, and -15 / -45 / -90 bp.

Scenario	Duration P&L (\$)	Duration + convexity P&L (\$)	Exact P&L (\$)
Parallel +50 bp	-624,435.38	-614,156.79	-614,300.03
Bear steepener	-613,244.75	-596,638.80	-597,026.60
Bull flattener	755,876.01	780,107.18	780,811.22

Scenario	Duration error (\$)	Combined error (\$)
Parallel +50 bp	-10,135.35	143.25
Bear steepener	-16,218.15	387.80
Bull flattener	-24,935.22	-704.05

Calculation and interpretation. Under +50 bp, portfolio duration P&L is $-50 \times \$12,488.707618 = \$-624,435.38$. The aggregate convexity correction adds \$10,278.59, giving \$-614,156.79. The signed errors are \$-10,135.35 for duration and \$143.25 for duration plus convexity. Exact repricing remains the benchmark because a finite yield move contains higher-order effects beyond the second derivative.

For the corporate bond alone, the +50 bp calculation starts from \$7,792,674.95, not its \$8 million face value. Multiplying by -7.478738×0.005 gives \$-291,396.89. Adding $\frac{1}{2} \times 7,792,674.95 \times 69.298064 \times 0.005^2$ adds \$6,750.22. Exact valuation uses 19 coupons of \$230,000 and a final \$8,230,000 discounted at 6.60%/2 per half-year. Their value is \$7,507,915.38, so the exact position P&L is \$-284,759.57.

Concentration. The corporate position is 26.22% of market value but 46.67% of DV01. Its later cash flows give it more sensitivity per invested dollar. In the bear steepener, its yield rises 75 bp, compared with only 10 bp for the Treasury; multiplying total DV01 by a simple average shock would ignore that concentration.

Proposed action. Sell \$1 million face of the corporate bond and hold the proceeds in cash assumed to have zero instantaneous price DV01. Each \$1 million face sale removes \$189.93/bp for the Treasury, \$438.17/bp for the agency, or \$728.49/bp for the corporate bond. The proposed sale reduces portfolio DV01 to \$11,760.22/bp, a 5.83% reduction. It avoids \$35,594.95 of the exact +50 bp loss. This is the most DV01 reduction per dollar of face sold among these three positions, not a claim that it is the best trade for every mandate. It also reduces corporate exposure and coupon income and changes the portfolio's maturity allocation.